

Appendices

Appendix A: Mathematical Formulations and Derivations

A.1 Reasoning Predictive Equity (RPE)

To quantify the semantic preservation of reasoning trajectories across successive generations, we define the Reasoning Predictive Equity (RPE). Let T_0 represent the baseline reference reasoning trajectory generated by the model at generation 0 (or derived from a verified gold standard), and let T_n represent the reasoning trajectory generated at recursive step n .

Each trajectory is represented as a sequence of semantic tokens or embeddings in a latent space. The similarity metric is formulated as:

$$\text{RPE}(n) = \cos(\mathbf{e}(T_n), \mathbf{e}(T_0)) = \frac{\mathbf{e}(T_n) \cdot \mathbf{e}(T_0)}{\|\mathbf{e}(T_n)\| \|\mathbf{e}(T_0)\|}$$

where $\mathbf{e}(T_i)$ denotes the pooled contextual embedding vector of the reasoning path extracted from the model's hidden states (typically the final layer activations before the projection head, or via an auxiliary encoder such as `text-embedding-3-large`).

A.2 The Stability Horizon (n_0)

The empirical decay of RPE over successive generations is modeled as an exponential decay process:

$$\text{RPE}(n) = \text{RPE}_0 \cdot e^{-k \cdot n}$$

where RPE_0 is the initial semantic alignment, and k is the model-specific fragility coefficient. To determine the operational boundary of coherent reasoning, we define the Stability Horizon (n_0) as the generation step at which

the alignment falls below a critical operational threshold RPE_{thr} (empirically set at 0.5):

$$\text{RPE}_{\text{thr}} = \text{RPE}_0 \cdot e^{-k \cdot n_0}$$

Solving for n_0 yields:

$$n_0 = \frac{1}{k} \ln \left(\frac{\text{RPE}_0}{\text{RPE}_{\text{thr}}} \right)$$

Appendix B: Experimental Setup and Hyperparameters

B.1 Model Implementations and Parameter Configurations

All evaluations were conducted using PyTorch and Hugging Face `transformers` libraries, distributed across an array of 8 NVIDIA H100 (80GB) GPUs. Table B.1 outlines the exact model versions, parameter scales, and generation hyperparameters utilized to ensure reproducibility.

Model Identifier	Parameter Count	Precision / Quantization	Temperature (T)	Top- p	Max Tokens
Llama-3-8B-Instruct	8 Billion	BF16	0.0 (Greedy)	1.0	512
Llama-3-70B-Instruct	70 Billion	BF16	0.0 (Greedy)	1.0	512
Qwen-2-7B-Instruct	7 Billion	BF16	0.0 (Greedy)	1.0	512
Qwen-2-72B-Instruct	72 Billion	BF16	0.0 (Greedy)	1.0	512
GPT-4o (2024-05-13)	Unknown	FP8 (via API)	0.0	1.0	1024
Claude-3-Opus	Unknown	FP8 (via API)	0.0	1.0	1024

Note: Greedy decoding ($T = 0$) was enforced across all evaluations to eliminate stochastic noise and isolate the deterministic drift caused strictly by autoregressive feedback loops.

B.2 Prompting Templates

To prevent prompt-induced biases, we utilized standard Zero-Shot Chain-of-Thought (CoT) system prompts.

System Prompt: You are an expert mathematical reasoning agent. Solve the user's problem step-by-step. Provide a clear, detailed, and logical chain of thought explaining your intermediate steps. End your response by clearly stating the final answer in the format: "Therefore, the final answer is: [ANSWER]".

User Input (Generation 0):

User Input (Generation n , where $n > 0$): Review the previous reasoning path and rewrite the step-by-step solution to ensure maximal logical consistency and mathematical accuracy. Keep the final answer identical. Previous path: {Output from Generation $n-1$ }

Appendix C: Bayesian Posterior Density Analysis

To confirm the statistical validity of the differences in fragility coefficients (k) between open-source and proprietary models, we performed a Bayesian linear regression using Markov Chain Monte Carlo (MCMC) sampling.

C.1 Model Specification

We defined a hierarchical model where the observed fragility k_i for model i on task t is modeled as:

$$k_i \sim \mathcal{N}(\mu_i, \sigma^2)$$
$$\mu_i = \beta_0 + \beta_{\text{scale}} \cdot \log_{10}(\text{Parameters}) + \beta_{\text{complexity}} \cdot \text{TaskDifficulty}$$

We assigned weakly informative priors to the coefficients: $\beta_0 \sim \mathcal{N}(0.2, 0.1)$, $\beta_{\text{scale}} \sim \mathcal{N}(-0.05, 0.02)$, $\beta_{\text{complexity}} \sim \mathcal{N}(0.1, 0.05)$, and $\sigma \sim \text{HalfNormal}(0.05)$.

C.2 MCMC Diagnostics and Posterior Intervals

Sampling was executed using the No-U-Turn Sampler (NUTS) with 4 chains, 2,000 tuning steps, and 2,000 draw steps per chain. All parameters achieved convergence with $\hat{R} < 1.01$. The 95% Highest Density Interval (HDI) for β_{scale} lies entirely below zero $[-0.072, -0.028]$, while the 95% HDI for $\beta_{\text{complexity}}$ is positively bounded $[0.045, 0.121]$.

Appendix D: Qualitative Case Studies of Collapse

Case Study D.1: Semantic Looping (GSM8K Task)

- Generation 0: "A farmer has 12 apples. He sells 4, leaving him with $12 - 4 = 8$ apples. He then buys twice as many as he sold, which is $2 \times 4 = 8$ apples. Total apples: $8 + 8 = 16$. Therefore, the final answer is: 16."
- Generation 9: "The farmer starts with 12. Sells 4. He has 8 remaining. Twice the amount sold is 8. He has 8. Twice the amount sold is 8. He has 8. This maintains the count. Twice of 8 is 16. Sells 4. He has 8. [Repeats indefinitely]"

Case Study D.2: Hallucinatory Drift (MATH Task)

- Generation 0: "Derivative of $f(x) = x^2 \ln(x)$. Using product rule, $f'(x) = 2x \ln(x) + x^2(1/x) = 2x \ln(x) + x$."
- Generation 12: "By definition of the logarithmic identity, the derivative must satisfy the identity boundary conditions. Thus, $f'(x)$ must represent the integral path of the function, which equates to $x^2 + \ln(x) + C$."